Prepare answers before the exercise class. In class you will present them.

1.1 Which group of properties is characteristic for any distributed system?

T F (T=true/F=false)

☐ ☐ Independent computers, connected by a communication network, synchronized clocks

☐ ☐ Independent computers, common shared memory, asynchronous communication network

☐ ☐ Messages-based communication, common task, common shared memory

☐ ☐ Multiple independent computers, no common clock, no shared memory

1.2 Which attributes are essential characteristics of a dependable distributed system?

T F (T=true/F=false)

☐ ☐ Reliability, Maintainability, Availability

☐ ☐ Portability, Scalability, Reliability

☐ ☐ Safety, Security, Integrity

☐ ☐ Timeliness, Availability, Cost-effectiveness

1.3 Dependability

a) In two or three sentences: dependability is not the same thing as reliability. What is the difference?

b) Name the three dependability attributes discussed with the measures MTTF, MTBF, and MTTR, and say which measure belongs to which attribute.

c) System A fails once per year and takes one day to repair. System B fails once per week and takes one hour to repair. Which system is more available?

1.4 Errors, failures, and faults

a) Put error, failure, and fault in causal order, and give a one-sentence definition of each.

b) A program in C uses a local variable that the programmer never initialized. The function is later called, and the program crashes. Identify the error, the failure, and the fault.

c) Give a second example of your own, from hardware or from a service you use. The example on the lecture slides (a stuck RAM bit) does not count.

1.5 How to get there

The lecture distinguishes fault prevention, fault removal, fault forecasting, and fault tolerance. Assign each measure to one of the four. One sentence of justification is enough.

a) Code review and testing before release.

b) Storing the same data on three disks and reading a majority.

c) Estimating how often a disk model fails, from field data.

d) Choosing a language in which an uninitialized variable does not compile.

1.6 Problems in Distributed Systems

a) In your opinion, what are the most important practical problems and challenges in distributed systems? You should especially focus on *technical problems* that should be solved by *algorithms*! Consider the experience you have from other lectures and projects, as well as outside of the university.

b) Which of the problems are especially interesting for you? For which of them do you want to learn more in this lecture?